Sahel Sharifymoghaddam

Ph.D. Student, IR/NLP
University of Waterloo

+1(647)771-3706
sahel.sharifymoghaddam@uwaterloo.ca
/sahelsharifymoghaddam
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Profile

PhD student in IR/NLP: LLM ranking eval, RAG grounding, agentic IR; multimodal.
Ex-Senior SWE, Google Cloud Applied AI.

Education

Ph.D. in Computer Science (University of Waterloo, Waterloo, Ontario, Canada) **2023 - present**

- Research Interests: IR for LLMs and LLMs for IR
- Supervisor: Prof. J. Lin (jimmylin@uwaterloo.ca)
- Coursework: Optimization for Data Science, Advanced Topics in AI (Trust and Explainablity), Advanced Topics in AI (Foundation Models), Advanced Topics in AI (Diffusion Models), **GPA: 95.5/100**

M.A.Sc. in Computer Engineering (University of Toronto, Toronto, Ontario, Canada) **2013 - 2015**

- Thesis: Optimized Data Placement for In-memory Data Analytics (A+)
- Supervisor: Prof. C. Amza (deceased), verification: Prof. N. Koudas (nick.koudas@utoronto.ca)
- Coursework: Advanced Operating Systems, Dependable Software Systems, Parallel Programming, Advanced Topics in Data Management Systems, Trends in Middle-ware Systems (Big Data), **GPA: 3.94/4**

B.Sc. in Computer Engineering (Sharif University of Technology, Tehran, Iran) **2009 - 2013**

- Selected Coursework: Object Oriented System Design, Data Structures and Algorithms, Design of Algorithms, Computer Architecture, Compiler Design, Computer Network, Operation Systems, Database Systems, Artificial Intelligence and eight courses on various Math topics, **GPA: 18.56/20** (Department average for class of 2013 is 16.03.)

Selected Publications

- Ge, Y.*, **Sharifymoghaddam, S.*** and Lin, J., Lighting the Way for BRIGHT: Reproducible Baselines with Anserini, Pyserini, and RankLLM. arXiv:2509.02558 (2025).
- Chen, Z., Ma, X., Zhuang, S., Nie, P., Zou, K., **Sharifymoghaddam, S.**, Liu, A., Green, J., Patel, K., Meng, R., Su, M., ... & Lin J. **BrowseComp-Plus: A More Fair and Transparent Evaluation Benchmark of Deep-Research Agent.** *(Spotlight acceptance at MTI-LLM NeurIPS 2025), arXiv:2508.06600 (2025).*
- **Sharifymoghaddam, S.**, Pradeep, R., Slavescu, A., Nguyen, R., Xu, A., Chen, Z., Zhang, Y., Chen, Y., Xian, J., & Lin, J. **RankLLM: A Python Package for Reranking with LLMs** *In Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2025)*, pp. 3681-3690
- **Sharifymoghaddam, S.***, Upadhyay, S.*, Thakur, N.*, Pradeep, R., & Lin, J. **Chatbot Arena Meets Nuggets: Towards Explanations and Diagnostics in the Evaluation of LLM Responses.** *arXiv:2504.20006. (2025)*
- **Sharifymoghaddam, S.***, Upadhyay, S.*, Chen, W., & Lin, J. **UniRAG: Universal Retrieval Augmentation for Large Vision Language Models.** *In Findings of the Association for Computational Linguistics: (NAACL 2025)*, pp. 2026-2039.
- Pradeep, R., Thakur, N., **Sharifymoghaddam, S.**, Zhang, E., Nguyen, R., Campos, D., ... & Lin, J. (2025). **Ragnarök: A reusable RAG framework and baselines for TREC 2024 retrieval-augmented generation track.** *In European Conference on Information Retrieval (ECIR 2025)*, pp. 132-148.
- Pradeep, R., **Sharifymoghaddam, S.**, & Lin, J. **RankZephyr: Effective and Robust Zero-Shot Listwise Reranking is a Breeze!** *arXiv:2312.02724 (2023)*
- Pradeep, R.*, **Sharifymoghaddam, S.***, & Lin, J. **RankVicuna: Zero-shot listwise document reranking with open-source large language models.** *arXiv:2309.15088 (2023)*

Research Experience

Ph.D. student **Sep. 2023 - Present**
University of Waterloo

- *LLM reranking and tooling*: Introduced **RankVicuna/RankZephyr** (listwise prompting, robust training) and built **RankLLM** (60k+ PyPi downloads, 540+ Github stars) with support for open/proprietary models, OpenRouter, thinking models, and more. Works have accrued 300+ citations.
- *RAG generation and evaluation*: Developed **UniRAG** to retrieve image–text pairs as in-context exemplars to guide LVLMs, improving faithfulness/quality in captioning and text-to-image; Contributed to **Ragnarök** to standardize the TREC 2024 RAG Track; and automated the **Search Arena evaluation using AutoNuggetizer** and found moderate alignment with human preferences.
- *Reasoning-intensive and agentic IR*: Co-developed **BrowseComp-Plus** benchmark to cleanly separates retriever vs. agent contributions for fair, transparent comparisons; In particular, I injected a reranking stage over retriever outputs before handing them to the orchestrator agent; I also released **reproducible BRIGHT baselines** with sparse/dense/fusion retirever + listwise rerankers using Pyserini, Anserini, and RankLLM.

M.A.Sc. student **Sep. 2013 - Aug. 2015**
University of Toronto

- *Thesis*: Developed a **Dynamic Vertical Partitioning (DVP) technique**, a workload-driven technique that groups sparse JSON attributes across in-memory tables via a cost model and greedy optimization to cut cache/TLB misses.
- *Additional project contributions*: Optimized **application placement for congestion/failure resiliency** in clouds; profiled **scheduling algorithms** for heterogeneous CPU–GPU clusters; and characterized Intel Haswell TSX hardware transactional memory, linking transaction size, write ratio, and retry counts to abort rates and performance.

Work Experience

AI Consultant (Part-time) **Aug. 2025 - Nov. 2025**
Yupp.ai

Worked on **open-sourcing a multi-turn image dataset generation and image-editing benchmark with human feedback**; responsibilities included dataset design, curation, and quality filtering; safety reviews and content moderation (NSFW, PII); spam/low-quality signal detection; and evaluation using VLM-as-a-judge. This work involved writing BigQuery SQLs for large-scale filtering/aggregation and developing Python notebooks for preprocessing, validation, and reproducible analyses.

Senior Machine Learning Engineer **Apr. 2021 - Apr. 2025**
Applied AI - Cloud - Google

Led multiple projects in the Sales AI and Healthcare AI teams, driving efforts across design, implementation, testing, and launch. Built and deployed **Generative AI applications on Google Cloud Platform (GCP)**, delivering end-to-end solutions for real-world enterprise use cases. Prior to working with Generative AI, focused on **fine-tuning domain-specific language models for core NLP tasks** such as Named Entity Recognition (NER), Relation Extraction, and Entity Linking/Disambiguation.

Selected Projects

- In Sales AI, worked on diverse data processing pipelines using Generative AI and prompt engineering techniques. Implemented **prompt optimization, LLM-as-a-judge evaluation, synthetic data generation, grounded generation with Google Search and citation insertion**, and ensured security and privacy compliance for the serving infrastructure.
- Led six feature launches for the Cloud Healthcare NLP API that processes medical notes for Entity Extraction, Relation Extraction, and Entity Linking. Responsibilities spanned design, implementation, and deployment on Vertex AI. Launches included integrating new Entity Extraction and Entity Linking models, and upgrading model serving infrastructure for performance and scalability.

- Developed GCP DocumentAI processors for Healthcare and Contract AI use cases. Trained custom entity extraction models, designed deployment architectures, scoped the projects into incremental, deliverable features, and led cross-functional execution involving nine engineers across three teams.

Software Engineer **Apr. 2016 - Apr. 2021**
Chromium Browser - Google

Worked end-to-end on multiple projects within the Web Payments and Input & Scrolling teams, leading efforts in design, implementation, launch, and public documentation. Actively guided cross-team collaborations, addressed high-priority feature requests, and contributed to W3C standards.

Selected Projects

- Led GPay's adoption of the Web Payments API, coordinating cross-team efforts, triaging feature requests, and analyzing the impact of Chrome's privacy sandbox on GPay APIs.
- Defined the Shipping Option and Address Change events in the W3C standards for the Payment Handler API, implemented them, and launched the features in Chromium—enabling real-time transaction updates for merchants.
- Improved scrolling smoothness and latency across all platforms by making document-level wheel event listeners passive by default and implementing scroll latching and asynchronous wheel events. Enhanced autoscroll latency and unified fling handling logic by migrating touchscreen, touchpad, and autoscroll fling events from the renderer process to the browser process, resolving multiple outstanding issues.

Selected Awards and Honors

- Recipient of the **Go-Bell** Scholarship (2023 - 2025), awarded by the University of Waterloos David R. Cheriton School of Computer Science to **a single incoming PhD student** among the Class of 2023.
- Recipient of the **Math Domestic Graduate** award and the **Dean of Math Excellence** scholarships awarded by University of Waterloo (2023 - 2024)
- Recipient of the **monthly scholarship for graduate studies** awarded by University of Toronto (2013-2015).
- **Achieved** 5th **highest GPA** among all computer engineering students (110+), Sharif University of Technology, class of 2013. Recipient of GPA-based **Honorary Admission for Graduate Study** (Declined).
- **Ranked** 4th in **National Scientific Olympiads** in Computer Engineering, Iran, 2012.

Selected Teaching Experience

- Teaching Assistant, University of Waterloo
  - **Information Systems Management**, with Prof. R. Cohen (Spring 2025)
  - **Computational Linguistics (graduate course)**, with Prof. F. Shi (Winter 2025)
  - **Object-Oriented Software Development**, with Caroline Kierstead (Fall 2025, Winter 2024, Fall 2023)
- Teaching Assistant, University of Toronto
  - **Design Algorithm**, with Dr. F. Pitt (Winter 2015)
  - **Foundations of Computing**, with Dr. D. C. Drosu (Winter 2015, Winter 2014)
  - **Parallel Programming (graduate course)**, with Prof. C. Amza (Fall 2014)

Computer Skills

- **Machine Learning:** Deep learning, Natural Language Processing, (multi-modal) LLMs, Diffusion Models, HuggingFace Transformers, TensorFlow and Keras, Pytorch, Pandas, Numpy, LangChain, LlamaIndex, Weights and Biases, vLLM, DeepSpeed
- **Programming Languages:** Proficient in Python, C++, Go, Java, and JavaScript; submitted and reviewed hundreds of thousands of lines of production code across 10+ languages at Google.
- **Google Cloud Basics:** Compute Engine, IAM, Spanner, BigQuery, Cloud Storage, Vertex AI, and Familiar with GCP Deployment Manager, Cloud Infrastructure, Docker and more.
- **Other Qualifications:** System Design, API Design (gRPC with protocol buffers, and W3C standards), Relational Databases (MySQL), Operating Systems, Network, Compilers, Computer Architecture, Data Structures and Algorithms, Git